Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Proxy Features and Fairness

> Prepare Proxy Features and Fairness: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Identify possible **proxy feature**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Subgroup Performance

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate subgroup results with sample sizes

Reference example, not the only acceptable wording: Teacher explanation: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Look for: Create features available at **prediction** time

Reference example, not the only acceptable wording: Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Record enough detail to reproduce a run

Reference example, not the only acceptable wording: Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Concept to explain

A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

Term	Meaning
proxy feature	input correlated with another sensitive attribute
fairness	evaluate benefits and harms across affected groups
paired test	cases differing in one relevant factor
mitigation	control reducing an identified risk

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **proxy feature**s and **fairness**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

No; examine errors, consequences, data and context.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain **proxy feature**s and **fairness** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Proxy Features and Fairness using the supplied case, labelled mechanism and

> reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Trace suspicious proxies and compare model behavior after a controlled removal.

MODEL IDEA A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Photo background predicts classroom location and is correlated with camera quality. Model uses background as a shortcut for health labels.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Photo background predicts classroom location and is correlated with camera quality.

Step 2

STEP 2 Model uses background as a shortcut for health labels.

Step 3

STEP 3 Create paired crops with the same leaf and different backgrounds; **prediction**s change.

Step 4

STEP 4 Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Background colour tracks collection location, associated with plant types.

The teaching case

Explain proxy risk and test.

Exact explanation to model

Why the result follows

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Explain the added worked example

Explain the added worked example · 1

An **input** can carry information about a characteristic without naming it. In a fictional school-device dataset, device code may be associated with building or access to newer equipment.

Removing a building field does not remove all related information. Do not infer a real student identity or protected characteristic in this exercise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **proxy feature**s and **fairness**.

Explain the added worked example · 2

Fictional records show device code X on 8 of 10 north-building examples and 2 of 10 south-building examples. Device and building are associated in this sample.

This does not prove device causes model errors or that every X device comes from north. The remaining cases contradict such a universal rule.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Proxy feature	A variable that can stand in for another characteristic
Correlation	An observed association, not proof of cause
Sensitive attribute	A characteristic requiring particular care in a context
Ablation	A comparison that removes a feature or component

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Proxy features and fairness at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
North examples	8 X of 10	2 other devices
South examples	2 X of 10	8 other devices
Review question	What does code represent?	Check context and errors

A feature may be legitimate for one purpose and inappropriate for another. Ask what is measured, when it is available, why it predicts and who might be disadvantaged. A correlation alone neither authorizes use nor proves harm. Talk through the intended decision and available alternatives.

Why might removing building but keeping device code leave related information? Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Proxy Features and Fairness

Removing an obvious setting field may leave another field that carries the same information.

The facts we will use

> **Supplied facts**
>
> Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/32

Proxy Features and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Remove direct field

The location column is absent from the new input table.



2. Check retained proxy

camera_code still identifies the setting perfectly in this supplied sample.



3. Audit effect

Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

CHANGE ONE THING

Camera codes are assigned independently of setting in a new balanced dataset.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Remove direct field	The location column is absent from the new input table.
2. Check retained proxy	camera_code still identifies the setting perfectly in this supplied sample.
3. Audit effect	Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Change one thing in the pictured case

Change one thing

Change: Camera codes are assigned independently of setting in a new balanced dataset.

Predicted result

Prediction: camera_code no longer perfectly identifies setting in that dataset.

Why the result follows

Reason: The observed mapping changed; other retained fields may still contain setting information.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Proxy Features and Fairness, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: Background colour tracks collection location, associated with plant types.

2. State the decision to explain

Write the question to explain: Explain proxy risk and test.

3. Reveal the worked reasoning

Reveal after **prediction**s: Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Hold back the independent task

Keep for an independent attempt: A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Device code is associated with building in the sample and may carry some of that information indirectly. Exit response: No.

Other features can retain information and accuracy can hide different errors and consequences.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Does equal overall accuracy prove fair treatment? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify proxy feature in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: read an association; find a proxy risk. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between proxy feature and correlation. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Describe a controlled with/without-code comparison.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Identify possible proxy features; Explain why removing one field does not guarantee fairness. Record any support used.

Stage: 35 to 45 minutes

> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Why might removing building but keeping device code leave related information?

Reasoning to listen for

Teacher check: Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

The retained G/D code maps exactly to greenhouse/desk, so the supplied table still exposes setting information. Changed-case check: camera_code no longer perfectly identifies setting in that dataset.

The observed mapping changed; other retained fields may still contain setting information.

Changed-case reasoning: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Proxy Features and Fairness investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Proxy Detective: map feature origins, run **paired test**s, remove one suspicious feature, and check both utility and gaps.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Removing a sensitive column guarantees the model cannot use related information.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

A stronger claim makes a stronger project: No. Other features can retain information and accuracy can hide different errors and consequences.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: The claim is unsupported: school code may be associated with location.

Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Proxy Features and Fairness questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Independent question

Independent question · withhold the key

A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> It may use camera artifacts as a proxy for species. Test across cameras with balanced species coverage and compare a model excluding the feature.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A feature unrelated to plant biology cannot carry label-correlated information.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Location may encode class prevalence rather than leaf **evidence**. Evaluate leaf classes across new and shared locations, inspect coverage and use appropriate grouping.

A correlated proxy can fail after deployment; high existing accuracy does not prove a useful causal feature.

Explanations for the additional workbook examples

W1 Read an association

Workbook question · W1

What do the device counts support, and what do they not support?

Teacher reasoning · W1

Teacher explanation: X occurs 8/10 north and 2/10 south in this sample. This supports an association, not a deterministic identity rule or a causal explanation.

W2 Find a proxy risk

Workbook question · W2

Why might removing building but keeping device code leave related information?

Teacher reasoning · W2

Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

W3 Plan an ablation

Workbook question · W3

Describe a controlled with/without-code comparison.

Teacher reasoning · W3

Teacher explanation: Use the same validation rows and metric, fit both versions only on **training**, and report overall and relevant subgroup errors with denominators.

W4 Apply independently

Workbook question · W4

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

Teacher reasoning · W4

Teacher explanation: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

W5 Limit the result

Workbook question · W5

Does unchanged accuracy after removing a proxy prove **fairness**?

Teacher reasoning · W5

Teacher explanation: No. Other features can retain information and accuracy can hide different errors and consequences.

Assess independent transfer

Independent question · withhold the key

Selected case: A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect and extend learning

> Connect Proxy Features and Fairness to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 33: Model Cards and Human Review. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **proxy feature** and **correlation** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Fresh retry: removing a device-brand field while retaining model number may leave brand-related information. Require a contextual association check and no automatic **fairness** claim.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Model cards and review. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/33, ai-4/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Proxy Features and Fairness lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Identify possible **proxy feature**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Why might removing building but keeping device code leave related information?

Supported practice answer

Reasoned practice answer: Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

Check every learner before discussion: Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Camera codes are assigned independently of setting in a new balanced dataset. Require a **prediction** and a reason before comparing. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

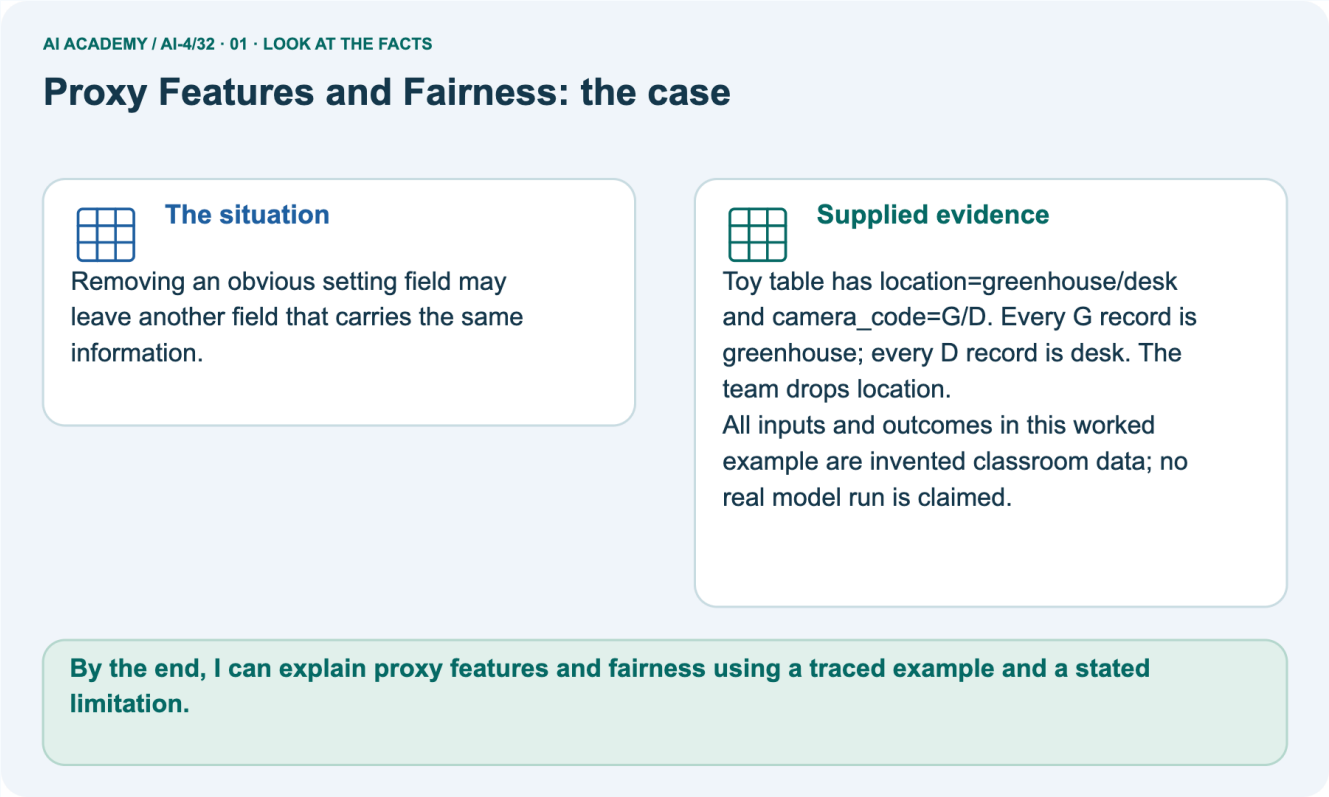
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Proxy Features and Fairness graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Proxy Features and Fairness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

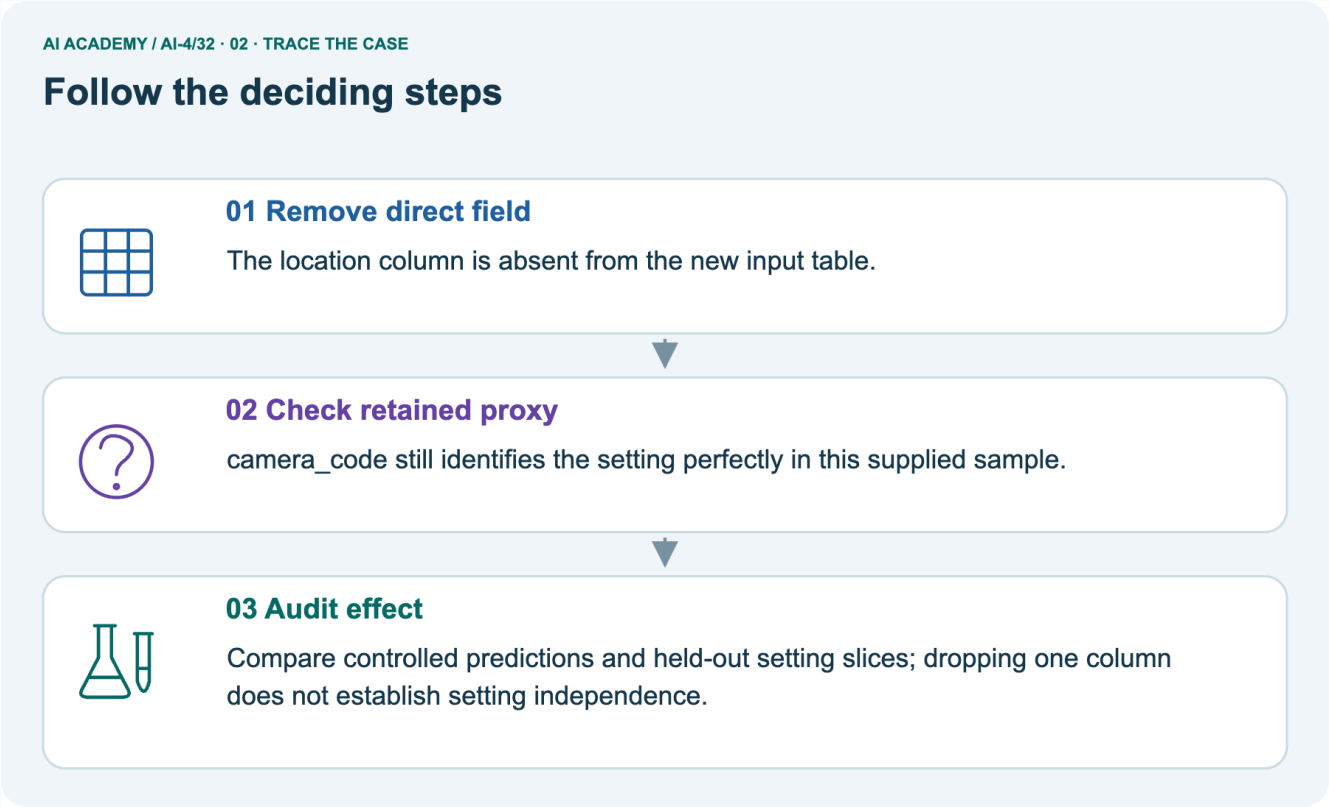
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

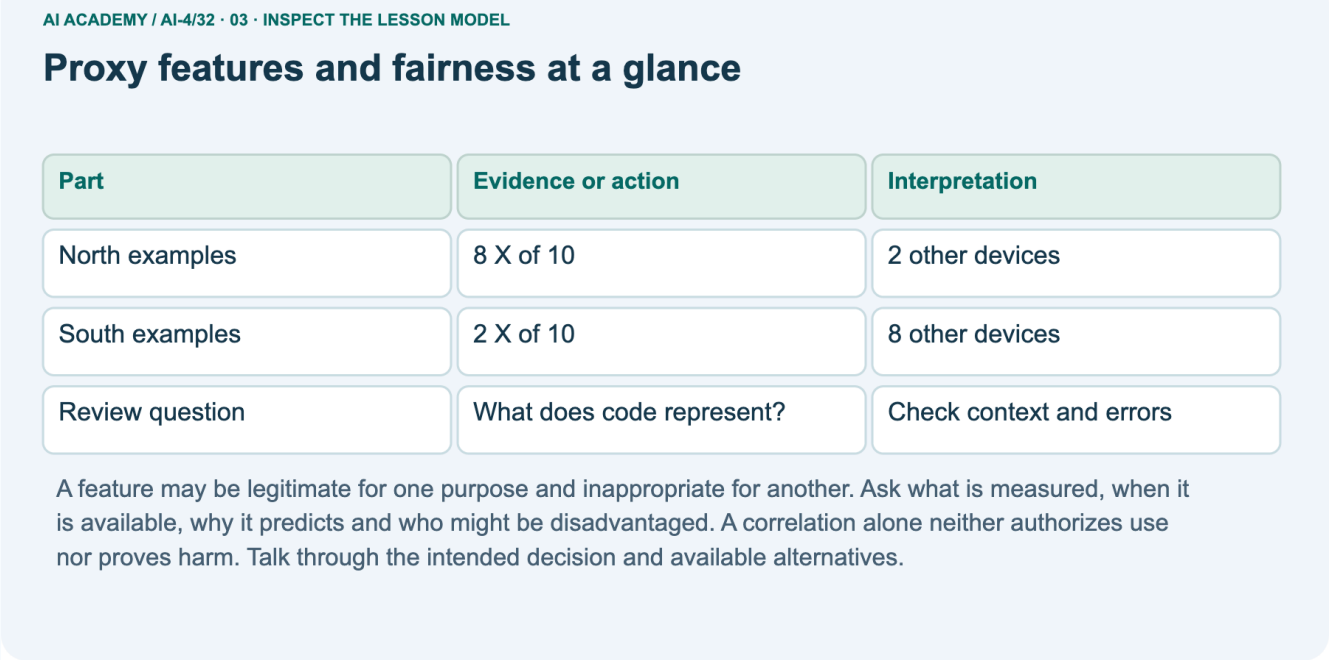
Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Proxy features and fairness at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

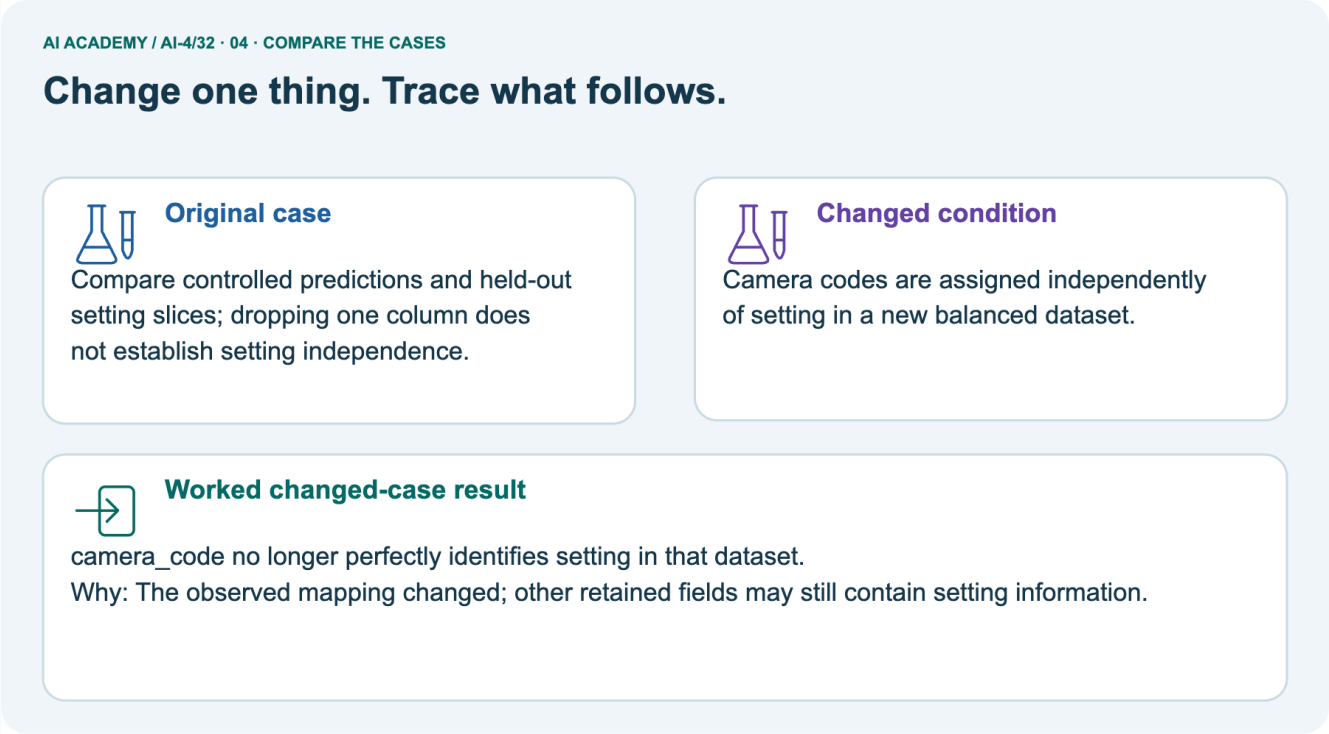
Proxy features and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

Predict before checking the worked changed-case result. Name what changed and what the

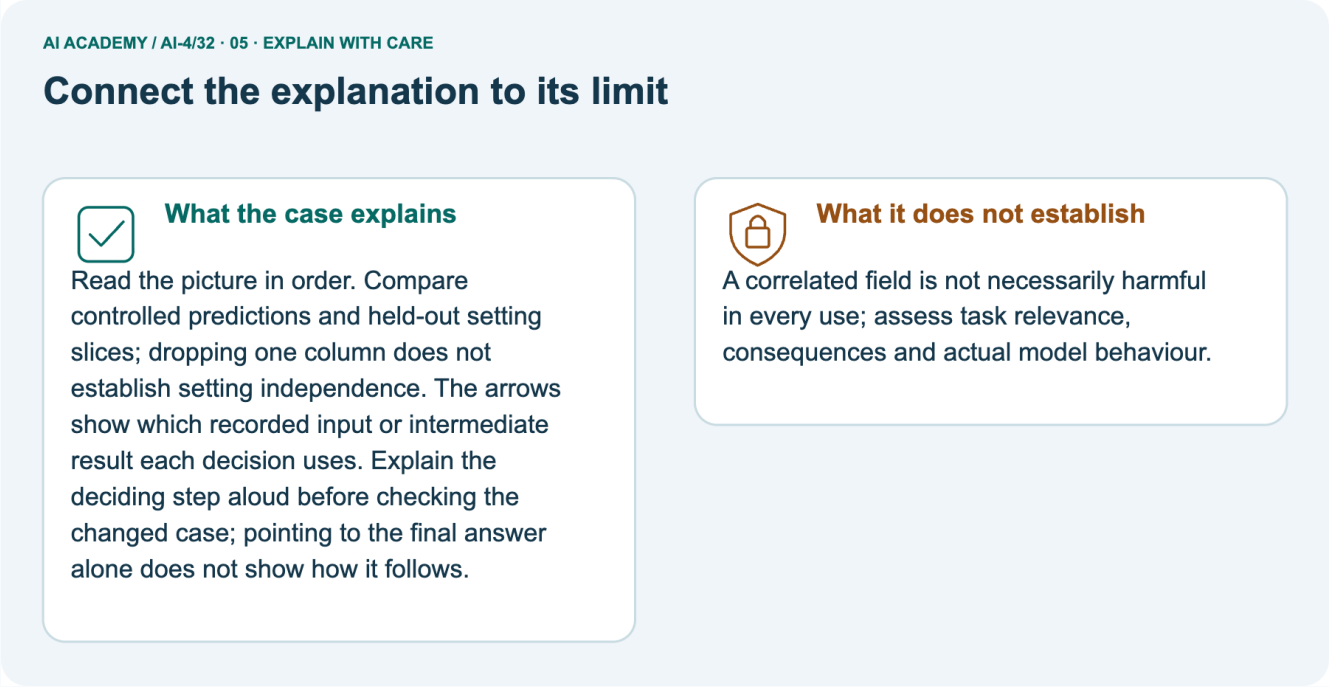
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The observed mapping changed; other retained fields may still contain setting information.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

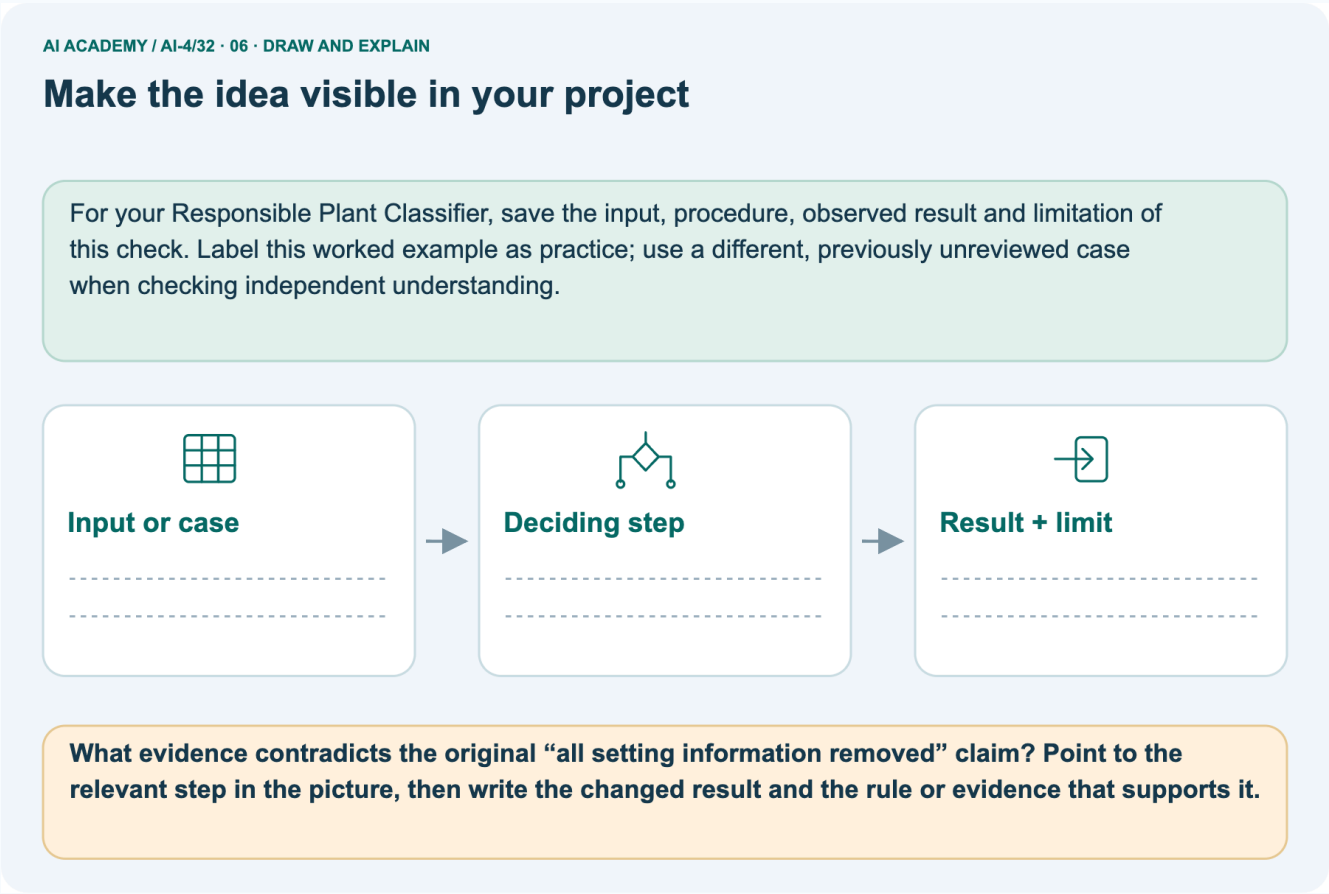
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The retained G/D code maps exactly to greenhouse/desk, so the supplied table still exposes setting information. Changed-case check: camera_code no longer perfectly identifies setting in that dataset.

The observed mapping changed; other retained fields may still contain setting information.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.